

The summer predictability barrier phenomena of symmetric and asymmetric Indian Ocean dipole events

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ABSTRACT

Predictions of Indian Ocean dipole (IOD) events often encounter the influence of the “summer predictability barrier” (SPB)—a phenomenon characterized by rapid prediction errors growth during the boreal summer. This study investigates the significant SPB associated with symmetric and asymmetric IOD events using the outputs of the Geophysical Fluid Dynamics Laboratory Earth System model. The results reveal that both types of IOD events experience the SPB regardless of the initial times of predictions. Furthermore, the significant SPB, which yields larger forecast error, is more likely to occur in the asymmetric IOD (asy-IOD) predictions when the predictions are initialized in October of the year preceding the IOD event. However, the predictions of symmetric IOD (sy-IOD) events with the initial time being January of the year that IOD events happen, are more inclined to experience the significant SPB. By analyzing the spatial characteristics of initial errors and tracking the evolutions of prediction errors, it is shown that the differences of the significant SPB for the asy- and sy-IOD events originate from the interactions between the inherent asymmetric of the IOD itself and the particular initial errors. This result may provide a theoretical basis for improvement of the IOD prediction skill using targeted observation or ensemble forecast techniques.

摘要

印度洋偶极子 (IOD) 在北半球夏季常出现预报误差急剧增大的现象, 该现象被称为夏季预报障碍 (SPB), 显著制约了IOD的预报能力. 本研究基于GFDL地球系统模式的预报数据, 深入探究了对称与非对称IOD事件SPB现象的误差增长动力学机制. 结果表明, 无论何时起报, 两类IOD事件均会出现SPB现象, 特别是当预报起始时间为IOD事件发生前一年的10月份时, 非对称IOD事件的预报更易出现显著的SPB; 而当预报起始时间为IOD事件发生当年的1月份时, 对称IOD事件的预报则更易产生显著的SPB. 通过对比分析, 发现两类IOD事件在不同起始月份更易出现显著SPB的主要原因源于IOD事件的不对称性特征与其更易引发SPB的特定初始误差的共同作用所致. 这一发现为通过目标观测或集合预报技术提升IOD预报水平提供了重要的理论依据.

1. Introduction

The Indian Ocean dipole (IOD), which presents positive and negative phases, is one of the most important interannual variabilities over the Indian Ocean (IO). A positive IOD (pIOD) event is manifested by a dipole mode with negative sea surface temperature anomalies (SSTAs) in the southeastern Indian Ocean (SEIO) and positive SSTAs in the western Indian Ocean (WIO), accompanied by equatorial easterly anomalies (Saji et al., 1999; Saji and Yamagata, 2003a), which not only affect the temperature and precipitation in the area around the IO (Saji et al., 1999; Saji and Yamagata, 2003b; Behera et al., 2005), but also influence further regions through planetary waves (Li and Mu, 2001;

Yan and Zhang, 2004; Saji and Yamagata, 2003b; Chan et al., 2008). However, the prediction skill for the IOD, as evaluated in previous research, is limited to three to four months (Doi et al., 2020; Song et al., 2022). Several studies have proposed that the winter predictability barrier and the summer predictability barrier (SPB) of IOD events may be responsible for such finite prediction skill (Wajsowicz, 2005; Luo et al., 2007; Zhao and Hendon, 2009; Feng and Duan, 2014a, 2014b; Liu et al., 2018).

Recent research has shown that the intensity of SSTAs related with pIOD events presents an extreme case, i.e., the amplitude of SSTAs in the SEIO is occasionally twice as strong as that in the WIO, defined as asymmetric IOD events (Drbohlav et al., 2007; Sun et al., 2014). The

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asymmetric feature of pIOD events not only directly influences the rainfall of India (Loschnigg et al., 2003), but also the precipitation of China (Zhang, 2001; Ding and Liu, 2001). Therefore, the different effects of asymmetric and symmetric IOD events make us naturally question the predictability of these two types of pIOD events and whether there are any differences in the predictability barrier phenomenon.

In order to answer these questions, we use a data analysis approach for predictability proposed by Hou et al. (2019), based on the concept of perfect model predictability experiments (Duan et al., 2009), to explore the predictability of asymmetric (asy-) and symmetric (sy-) IOD events.

2. Data and methods

The outputs of the pre-industrial (PI) control run within the phase 6 of the Coupled Model Intercomparison Project (CMIP6) are employed to evaluate the predictability of asymmetric and symmetric IOD events. The PI-Control run is a long-term run of the model with the external forcing, such as the greenhouse gas concentrations, being fixed at 1850, which excludes the influence of varied external forcing and only contains the internal variations of the models. Simulations of the IOD have been examined in the 34 CMIP6 models, among which the Geophysical Fluid Dynamics Laboratory Earth System (GFDL-ESM4) presents well performance (McKenna et al., 2020). In this research, the 100-year monthly averaged sea temperature and sea surface wind are used. Details of GFDL-ESM4 can be found in Krasting et al. (2018).

To study the predictability of asy-iod and sy-iod, we employ a data analysis approach for predictability proposed by Hou et al. (2019), which is appropriate for variables with a diurnal or monthly cycle. As this work mainly focuses on the differences in the predictability barrier considering the asymmetry of IOD events, to avoid other influencing factors, especially El Niño–Southern Oscillation (ENSO), which has been demonstrated to present significant effects on the SPB of the IOD (Liu et al., 2021), five “pure” asy-iod and sy-iod events, which did not

coincide with ENSO, are randomly chosen from the 100-year outputs according to the amplitude of SSTAs over the WIO and SEIO (Fig. S1). The evolutions of each IOD event within one year are considered to be the real “observation”, and the 15 years prior to and following the referenced IOD events are assumed to be 30 “predictions” with a lead time of 12 months to the observation. Upon this basis, the differences between the initial states of these two 12-month slices are considered to be initial errors, and the prediction errors are defined as the absolute values of the deviations between the “predicted” and “observed” dipole mode index (DMI), which is the difference in SSTAs between the WIO and the SEIO and usually used to measure the occurrence and intensity of the IOD. Then, the growth rate of prediction errors is calculated as follows:

$$\kappa = \frac{\partial P(t)}{\partial t} \approx \frac{P(t_0 + \Delta t) - P(t_0)}{\Delta t}, \quad (1)$$

where $P(t)$ represents the prediction errors at time t , t_0 is the initial time. As the monthly outputs being used in this research, the Δt is one month and κ indicates the monthly variation of prediction errors. To explore the seasonal variability of the prediction errors, the seasonal mean is calculated by taking the average of the κ during three consecutive months.

3. Results and discussion

3.1. The predictability barrier of symmetric and asymmetric IOD events

As previous studies have shown that the predictability barrier during the growth phase of the IOD is much stronger (Feng and Duan, 2014a; Liu et al., 2018), we calculate the moving average with one month intervals of three consecutive months with the start month being the previous October before the IOD year (October(−1)) and the January in the IOD year (January(0)) to investigate the seasonal evolutions of prediction errors of sy-iod and asy-iod events during their growth phases. Fig. 1

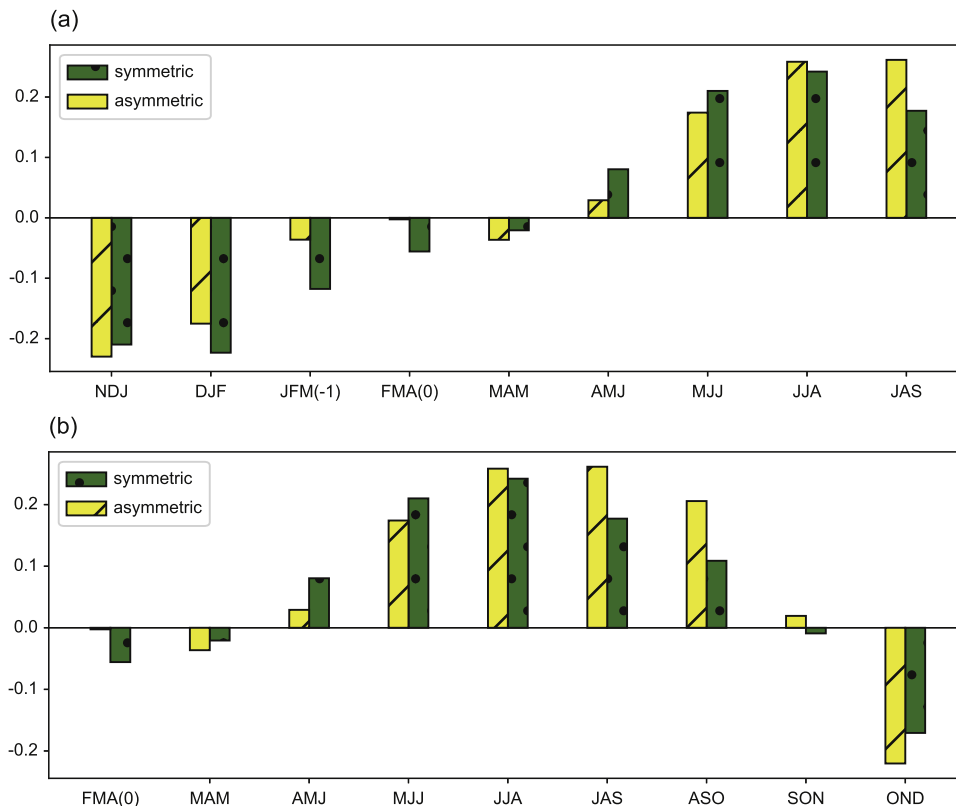


Fig. 1. Histograms of the seasonal growth rate κ for prediction errors of symmetric IOD (green bars) and asymmetric IOD (yellow bars), calculated by taking the ensemble mean of 150 predictions for each kind IOD event, with the start month being (a) October(−1) and (b) January(0).

Table 1 The numbers of members with an SPB according to different multiples of standard deviations.

Start month	Multiples of standard deviation	1	1.2	1.5	1.8	2.0	2.5
Jan(0)	Symmetric	14	13	10	9	7	4
	Asymmetric	12	11	10	9	7	3
Oct(-1)	Symmetric	12	11	9	7	7	4
	Asymmetric	18	15	12	11	10	7

shows the ensemble mean of the seasonal growth rate of 150 prediction errors for each kind of IOD event at two different initial times. The results show that the prediction errors tend to grow fastest during summer, from June to August, denoted by JJA(0), for sy-IOD, and July to September (JAS(0)) for asy-IOD, regardless of the initial time, which indicates the occurrence of the SPB in symmetric and asymmetric IOD predictions.

By evaluating the seasonal error growth rates of 150 predictions for sy-IOD or asy-IOD events, the SPB is found in both kinds of IOD event. In order to further explore the SPB considering the effects of IOD symmetry, we focus on the IOD predictions that experienced an SPB and ultimately yielded large prediction errors at the end time. The top five members are selected according to the ratio between the prediction errors at the initial and end time, which ensure the increasing of initial errors. Thus, we obtain 25 members for each kind of IOD event with the start month being October(-1) and January(0), respectively. Liu et al. (2018) sorted IOD predictions using the difference between the largest growth rate in summer and the second largest one, and defined a significant SPB is one larger than 1.5 standard deviations of these deviations. Accordingly, the amplitude of the SPB can be studied through the multiples of such standard deviation. The amounts of members selected by growing multiples are shown in Table 1. Comparing the results at two start months, we find that a significant SPB tends to appear more frequently in sy-IOD predictions with the initial time being January(0). That is, sy-IOD predictions tend to experience a stronger SPB when starting from January(0). However, when the predictions start from October(-1), there is a greater tendency for an asy-IOD event to experience an intense SPB, as the quantity of members that experience an SPB is consistently larger than that of the sy-IOD events. Considering the symmetry of IOD events, the frequency of SPB is different at two initial times. The SPB implies rapid growth of prediction errors, which will eventually lead to underestimation of the predicted IOD (Liu et al., 2021); and the stronger the SPB, the larger the predicted uncertainties. Thus, in this study, we mainly focus on the reasons for the frequent occurrence of a strong SPB for the sy- and asy- IOD events starting from two initial times.

3.2. Reasons for the occurrence of a significant SPB

The SPB for IOD predictions has been investigated by Liu et al. (2018), who showed that the initial errors over the Pacific Ocean (PO) play a key role in the appearance of the SPB. Such initial errors lead to anomalous westerly winds over the SEIO, which are greatly suppressed by the climatological and IOD-related winds during summer, and then reduce the loss of latent heat flux, favoring the growth of SST errors and eventually yielding an SPB. As mentioned in Section 2, the prediction errors in this research are only induced by the initial errors, so we first explore the spatial patterns of the initial errors using combined empirical orthogonal function (CEOF) analysis. The CEOF analysis is performed on the SST and subsurface temperature errors over the PO and IO of the 25 members for each kind of IOD event, which cover 10°N to 10°S and 50°E to 85°W from the surface to a depth of 200 m. Due to the first mode (CEOF1) explaining more than 30 % of the total variance, which is double that of the second mode, CEOF1 can represent the main spatial features of all the initial errors that contribute to the occurrence of the SPB. Moreover, the related principle component of CEOF1 (PC1) presents positive and negative values, which means some of these

members named by type-1 share a similar pattern with CEOF1, while the others may be almost opposite (hereafter known as type-2). Hence, we take the composite of these two types of initial errors to explore the spatial characteristics.

3.2.1. Reasons for the occurrence of a significant SPB for asymmetric IOD

Since the prediction errors are all induced by the initial errors, the reasons behind the more frequently appearance of a significant SPB for the asy-IOD predictions with the start time being October(-1) should be investigated by comparing the differences in the initial errors and the dynamic mechanisms related with error growth. For the asy-IOD and sy-IOD events, the large values of both types of initial errors are mostly concentrated over the PO, and the type-1 initial errors related with the two kinds of IOD events express negative SST errors over the eastern PO and a dipole mode of the subsurface sea temperature, with positive anomalies in the upper eastern PO and negative values in the lower western PO. The second type of initial errors are almost opposite to the first type (Figs. S2 and S3). By imposing such initial errors on the initial field, it is found that, eventually, both of them exhibit fast development of positive SST errors over the SEIO, which is proven to be responsible for the occurrence of an SPB (Figs. S4 and S5; Liu et al., 2018).

In order to explain why the asy-IOD frequently experiences a significant SPB, we further compare the seasonal growth rates of the WIO and SEIO for the sy- and asy-IOD events, as shown in Fig. 2(a). Regardless of the IOD type, the development of SST errors over the WIO maintains slightly and the growth of SST errors over the SEIO is extremely fast during summer (the lines and histograms are larger during July to September). Notably, the predicted SST errors and their growth rate in the SEIO for asy-IOD are both higher, which may determine the higher frequency of a significant SPB for asy-IOD predictions starting from October(-1). Actually, the changes in SST errors could be caused by the combined effects of the climatology, IOD events, and the initial errors (Liu et al., 2018), but the climatology is the same for sy- and asy-IOD in this study, and therefore the interaction between the IOD event itself and initial errors may contribute greatly to the different frequency of a significant SPB.

Previous studies indicate that the occurrence of equatorial easterly wind anomalies (EEWA) over the equatorial IO could impact the symmetry of IOD events (Drbohlav et al., 2007; Hong et al., 2008), with an earlier onset and sustained EEWA leading to stronger cooling over the SEIO; and the colder SST, which is transported by the westward ocean current, would suppress the development of positive SSTAs over the WIO, leading to the occurrence of asy-IOD (Sun et al., 2014). As shown in Fig. 3, we find an earlier occurrence of persistent stronger EEWA for the asy-IOD events starting from June, and in the meantime, initial errors generate anomalous westerlies, especially over the SEIO (Figs. S4 and S5), so the EEWA related with asy-IOD events could inhibit such westerly anomalies. It has been reported that variations of latent heat flux are positively related with wind speed (Cayan, 1992), and the wind anomalies over the SEIO induced by initial errors are greatly suppressed by the stronger EEWA during summer, decreasing the loss of latent heat flux, which is favorable for the rapid growth of positive SST errors. As a consequence, in comparison with the sy-IOD events, a significant SPB is usually presented in the asy-IOD predictions when starting from October(-1).

3.2.2. Reasons for the occurrence of a significant SPB for symmetric IOD

As the results in the above section showed, a prominent SPB tending to occur in the predictions of asy-IOD with the initial time being October(-1) is closely related with the mechanism that is also responsible for the occurrence of asy-IOD—specifically, the early appearance of IOD-related easterly anomalies over the equatorial IO, which considerably suppresses the westerlies caused by initial errors during summer. However, when the predictions start from January(0), there is a greater tendency for sy-IOD to experience a significant SPB, as shown in Table 1.

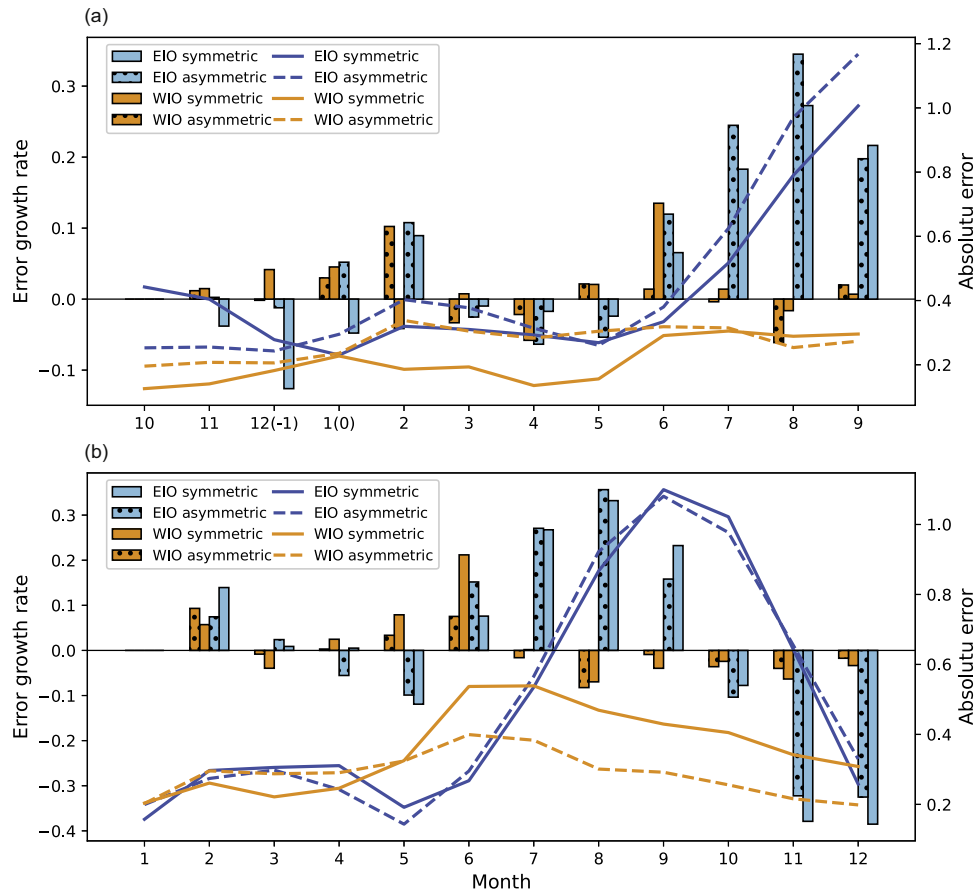


Fig. 2. Temporal evolutions of SST errors (lines; units: $^{\circ}\text{C}$) and the growth rates (histograms) of the SEIO (blue) and WIO (yellow) with the start month being (a) October(-1) and (b) January(0).

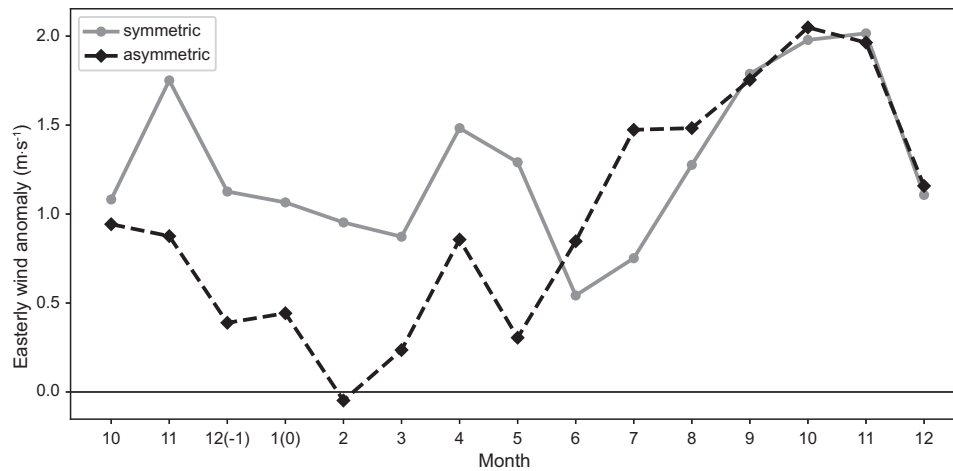


Fig. 3. Monthly evolutions of the regionally averaged (5°S–5°N, 70°–90°E) EEWA (units: m s^{-1}) of the two types of IOD, where the black dashed line represents the asymmetric IOD and the gray solid line represents the symmetric IOD.

Similar to the former analysis, two types of initial errors are also identified by using CEOF analysis, and the spatial distributions are quite similar to those initiated at October(-1), with much larger anomalies over the subsurface sea temperature in the IO (Figs. S6 and S7). The evolutions of these two types of initial errors present positive SST errors growing over the SEIO for both kinds of IOD, while negative SST errors over the WIO also evolve for the sy-IOD (Figs. S8 and S9).

For the predictions starting from January(0), the rate of SST errors over the WIO for the sy-IOD is larger than that of the asy-IOD during

summer, which may cause the more frequent appearance of a significant SPB (Fig. 2(b)). Further comparing the monthly development of SST errors over the IO for both kinds of IOD events, and taking the type-2 initial errors as an example, the positive SST errors over the SEIO progress rapidly from July to September, then reach a peak in October, but the negative SST errors over the WIO presented in the sy-IOD events (Fig. S10) do not occur in the asy-IOD events (Fig. S11). The SSTAs over the WIO are found to be possibly driven by the PO through the gear effect of double-cell Walker circulation (Chen, 2011). In this study, we also

found that the sustainable negative SST errors over the PO induced continuous easterly anomalies till August over the PO for the sy-IOD events; then, through the influence of Walker circulation, anomalous westerlies over the IO are maintained. Such westerlies will modulate the eastward ocean current through the air–sea interaction within the IO and cause upwelling over the WIO. Such upwelling would bring the locally amplified negative subsurface sea temperature errors to the surface and further enhance the sustainable development of negative SST errors over the WIO (Fig. S8), subsequently favoring the growth of negative SST errors over the WIO and ultimately inducing a frequently significant SPB for sy-IOD predictions. However, for the asy-IOD events, during the evolutions of prediction errors, due to the weak negative SST errors over the PO, the anomalous westerlies are restricted over the SEIO and the amplitude of SST errors over the WIO is weak and almost unchanged (Fig. S11). Therefore, the fast development of SST errors over the WIO is instrumental in causing a significant SPB for sy-IOD predictions when starting from January(0).

4. Conclusions

In this study, the predictability barrier for asymmetric (asy-) and symmetric (sy-) IOD events is investigated by using the outputs of GFDL-ESM4. From the perspective of initial error growth, by employing the data analysis approach for predictability introduced by Hou et al. (2019), the results showed that asy-IOD tends to experience a significant SPB with the initial time being October(−1), while an SPB usually occurs in sy-IOD when predictions start from January(0). We further explored the reasons for the different probability of the appearance of the SPB for asy- and sy-IOD, separately, and found that the interaction between the mechanisms that lead to the asymmetry of the IOD and the specific initial errors is also responsible for the difference in the SPB.

The spatial characteristics of two types of initial errors that cause SST errors to develop rapidly during summer and yield large prediction errors are identified, with large anomalies almost concentrated over the PO (Figs. S2–S9). By imposing such initial errors on the initial field and tracking the evolutions of prediction errors, the fast growth of positive SST errors over the SEIO during summer contribute to the frequent occurrence of the SPB for asy-IOD predictions starting from October(−1). The asy-IOD-related sustainable anomalous southeasterlies could restrain the anomalous westerlies caused by initial errors in summer, which prohibit the loss of the latent heat flux over the SEIO and favor the rapid development of positive SST errors at that time. Thus, the SPB ordinarily presents in asy-IOD predictions starting from October(−1).

For the prominent SPB in sy-IOD predictions with the initial time being January(0), the spatial distributions of initial errors are similar to those initiated at October(−1) but with much stronger SST errors over the IO. During the growth of such initial errors, the SEIO of sy-IOD is dominated by the positive SST errors and the negative SST errors over the WIO. For the asy-IOD, meanwhile, the negative SST errors are not shown. Comparing the evolutions of prediction errors, for the sy-IOD predictions, due to the influence of persistent negative SST errors over the PO, easterly anomalies are induced over the equatorial PO; through the gear effect of double-cell Walker circulation, anomalous westerlies are driven over the equatorial IO. Compared to asy-IOD events, such remarkable westerlies would modulate the eastward ocean current, with negative SST errors growing fast over the WIO via upwelling during summer, which could be conducive to a significant SPB occurring more easily.

In this study, considering the asymmetry of IOD events, the predictability barrier of IOD predictions can be driven by specific spatially characterized initial errors. This may emphasize the accuracy of the initial field on the improvement of IOD predictions, which has also been mentioned in previous research (Feng and Duan, 2014a; Liu et al., 2018). Furthermore, the large values of the initial errors for the sy- and asy-IOD predictions are concentrated in common areas, and this may

imply that preferentially assimilating additional observations in these areas and reducing the initial errors there can effectively decrease the SPB effect, eventually improving the prediction skill for both sy- and asy-IOD events. Therefore, even if one does not know which type of IOD event will occur, one can still increase its prediction skill through the above approach. This argument is related to the concept of “targeted observation”, which is aimed at improving the prediction skill through adding observations in the limited sensitive regions. There is no doubt that more experiments need to be done to certify whether the above-mentioned assumptions are effective for improving the IOD prediction skill.

Liu et al. (2018) showed that IOD predictions always experience an SPB due to combined effects of the particular initial errors, the climatological annual cycle, and the IOD events themselves. In this work, we found the spatial characteristics of SPB-related initial errors for the two types of IOD share great similarities. However, as illustrated above, the predictions of the two types of IOD events show different SPB frequencies depending on the start month. Therefore, these predictions are initialized under different phases of the climatological annual cycle and involve different IOD events. How, then, do different climatological annual cycle states and different IOD events interact with the similar initial errors, and then cause different frequencies of significant SPB occurrence? The answer to this question remains a challenge and needs to be explored in the future.

The former initial errors are considered to cause a significant SPB during the growth phase of the IOD, while Feng and Duan (2014a) found that there also exists a predictability barrier during the decaying phase of the IOD. Thus, more research is needed to analyze the predictability barrier for both types of IOD events during the decaying phase. Furthermore, to focus on the predictability barrier of sy- and asy-IOD events, ENSO was not considered in this research, as the effects of ENSO on the prediction skill for the IOD and the SPB have been reported in previous studies (Liu et al., 2021). How ENSO influences predictability barriers, especially the SPB, of the two kinds of IOD, should be explored in future studies.

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Supplementary materials

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